

TECNOLÓGICO NACIONAL DE MÉXICO

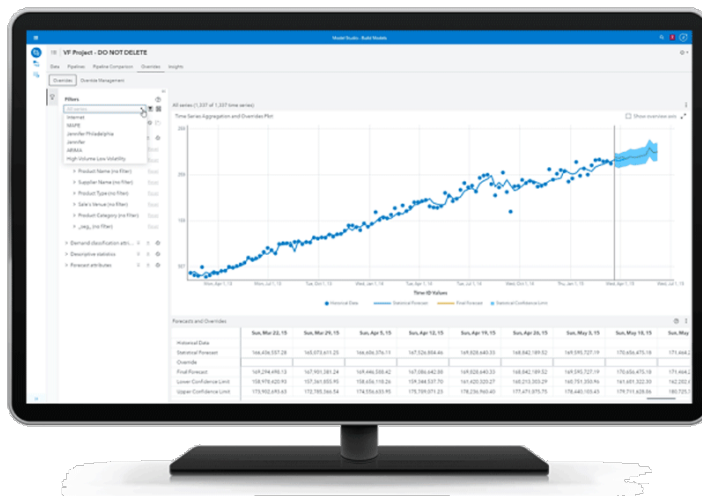
INSTITUTO TECNOLÓGICO DE CIUDAD GUZMÁN

Acondicionadores para Fuentes Renovables de Energía

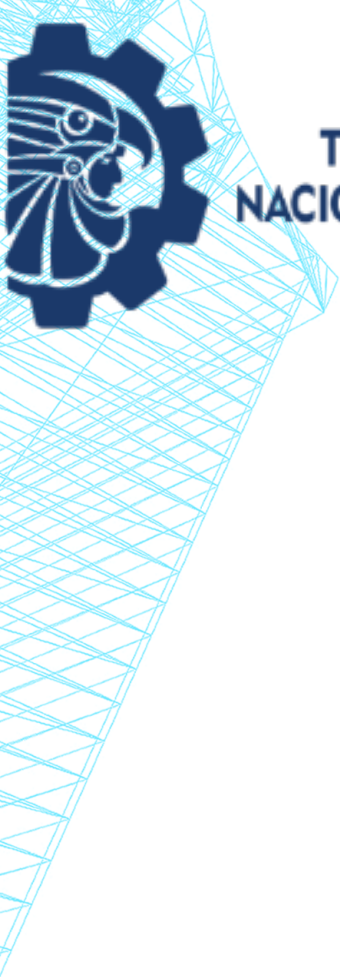
Maestría en Ingeniería Electrónica 3er Semestre

Presenta

Dr. Julio Alberto García Rodríguez



Ciudad Guzmán, Agosto de 2026



TECNOLÓGICO
NACIONAL DE MÉXICO

SITIO OFICIAL

<https://cdguzman.tecnm.mx/pag/>

EDUCACIÓN

SECRETARÍA DE EDUCACIÓN PÚBLICA



The screenshot shows the official website of TecNM Ciudad Guzmán. The header includes the logos of the Mexican Government, the Ministry of Education, and TecNM. Navigation links for 'Trámites', 'Gobierno', and a search icon are present. A secondary navigation bar lists various services like 'Estudiantes', 'Egresados', 'Avisos', 'Estadística', 'Transparencia', and 'Protección de Datos Personales'. Below this, a horizontal menu offers options such as 'Conoce', 'Nuevo Ingreso', 'Oferta Educativa', 'Eventos y Reconocimientos', 'Galería', 'ITCG', 'Recursos Humanos', and 'Reglamentos'. The main banner features a large blue arrow pointing right, followed by the text 'ESTUDIA UNA MAESTRÍA' in white, with 'MAESTRÍA' highlighted in a red speech bubble. Below this, a red box lists the master's programs: 'Maestría en: Ingeniería Electrónica (MIE)' and 'Ciencias de la Computación (MCC)'. To the right, a dark blue box says 'En el TecNM Ciudad Guzmán'. At the bottom, a link 'Requisitos y más información' is accompanied by a red button with a hand icon and the text 'Clic aquí'.

Gobierno de México

Trámites Gobierno

Estudiantes Egresados Avisos Estadística Transparencia Protección de Datos Personales

Conoce Nuevo Ingreso Oferta Educativa Eventos y Reconocimientos Galería ITCG Recursos Humanos Reglamentos

ESTUDIA UNA MAESTRÍA

Maestría en:

- Ingeniería Electrónica (MIE)
- Ciencias de la Computación (MCC)

En el TecNM Ciudad Guzmán

Requisitos y más información [Clic aquí](#)

ENCUADRE DE LA MATERIA

- **Actividades de aprendizaje** **30%**
(Tareas, Ejercicios de clase, Investigaciones, Prácticas, Exposiciones)
- **Instrumento de Evaluación: Examen escrito** **40%**
- **Proyecto de Asignatura** **30%**

A través del presente escrito CONFIRMO que conozco el ENCUADRE y el PROGRAMA DE ESTUDIOS de la materia de ACONDICIONADORES PARA FUENTES RENOVABLES DE ENERGÍA

Nombre completo y firma

Fecha

DATOS DE CONTACTO

Dr. Julio Alberto García Rodríguez

Correo: julio.gr@cdguzman.tecnm.mx

Oficina: Edificio Ñ, Cubiculo Jefatura de Ciencias básicas.

341-110-8223

CLASS BEST PRACTICES

- Limited Smartphone use. Emergencies only.
- Put attention in class, avoid distractors as Social Networks.
- Ask when you feel lost.
- You should meet the prerequisites. (Math, Programming, Soft skills).
- All resources must be in English.
- Some classes will be conducted entirely in English by both the instructor and the students.

Class Best practices

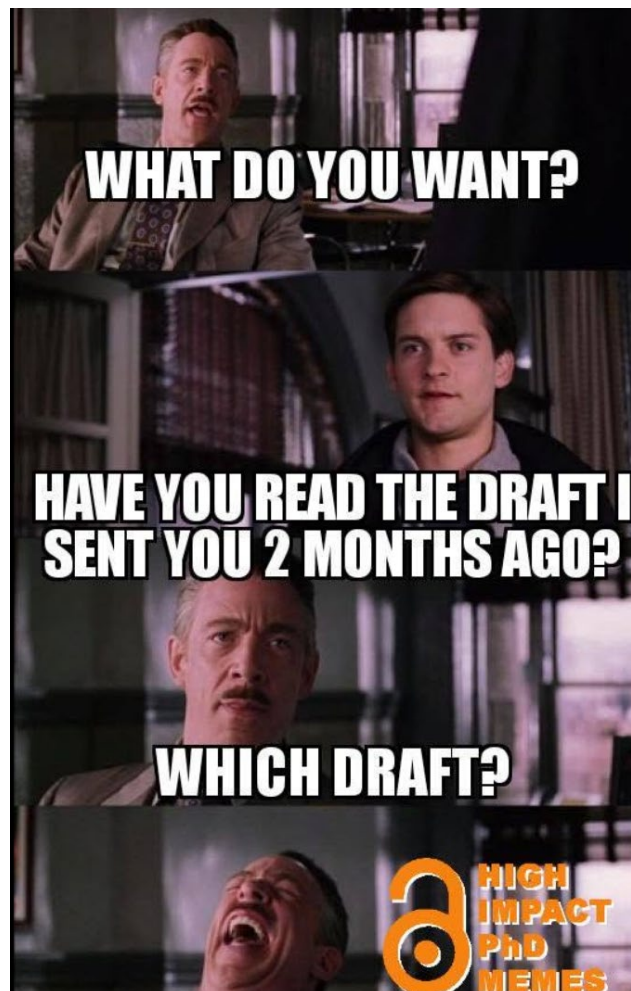
- Remember that Natural Intelligence is still better than Artificial Intelligence. Use AI tools wisely, responsibly, and ethically.
- The class focuses on developing logical, mathematical, and computational thinking skills. It is not focused on developing thousands of lines of code that even the student doesn't understand.
- Get your hands dirty.
- Stay engaged.

Class Best practices

Me sacrificing my sleep, sanity and soul for
my research,

Supervisor: This is not enough, you need to
work harder 😊

Me-



1. Photovoltaic Systems

1.1 Introduction

Global Warming

What is global warming?



1. Photovoltaic Systems

1.1 Introduction

Global Warming

Global warming is the gradual increase in average surface temperature over the decades.

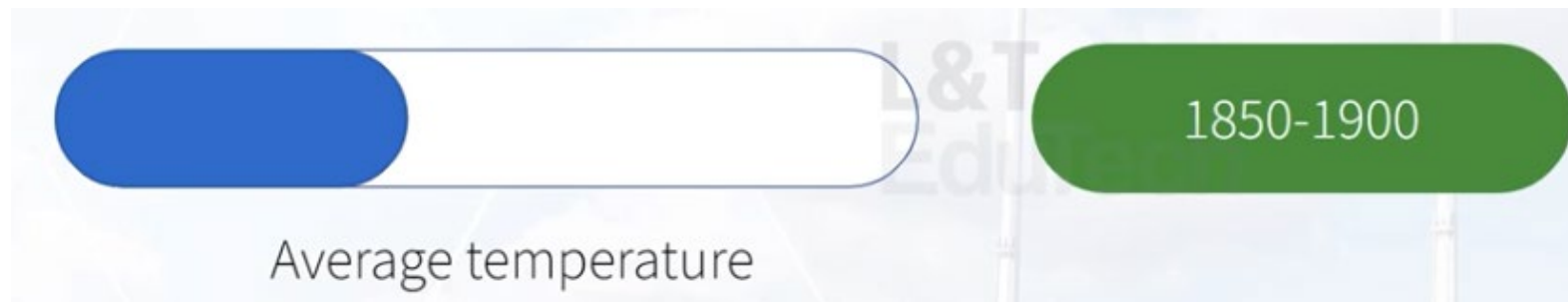


1. Photovoltaic Systems

1.1 Introduction

Global Warming

The increase in temperature has been gradual over the century. However, the measure of increase has to have a reference point, which in this case, is the average temperature that prevailed between 1850 and 1900.



The average temperature between 1850 and 1900 is about 13.84 °C (56.91 °F), which is the pre-industrial baseline at which the anomaly is zero.

1. Photovoltaic Systems

1.1 Introduction

Global Warming

Natural changes have contributed to the increase in global temperature. The biggest contributing factor has been the boom in industrial activity, specifically between 1850s and 1900s



- <https://www.climate.gov/news-features/understanding-climate/climate-change-global-temperature>
- <https://wmo.int/resources/dashboards/global-mean-temperature-1850-2024>
- <https://berkeleyearth.org/global-temperature-report-for-2025/>
- <https://climatechangetracker.org/global-warming/yearly-average-temperature>

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1.1 Introduction

Global Warming

The abundant use of fossil fuels in steam engines, petrol engines, diesel engines, and locomotives, followed by large scale industrial activity has resulted in significant negative impact on the environment.



Steam Engines



Petrol and Diesel Engines



Locomotives



Significant negative impact on the environment

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1.1 Introduction

Global Warming

Natural consequences, social threats, threats to business, Territorial threats, etc.



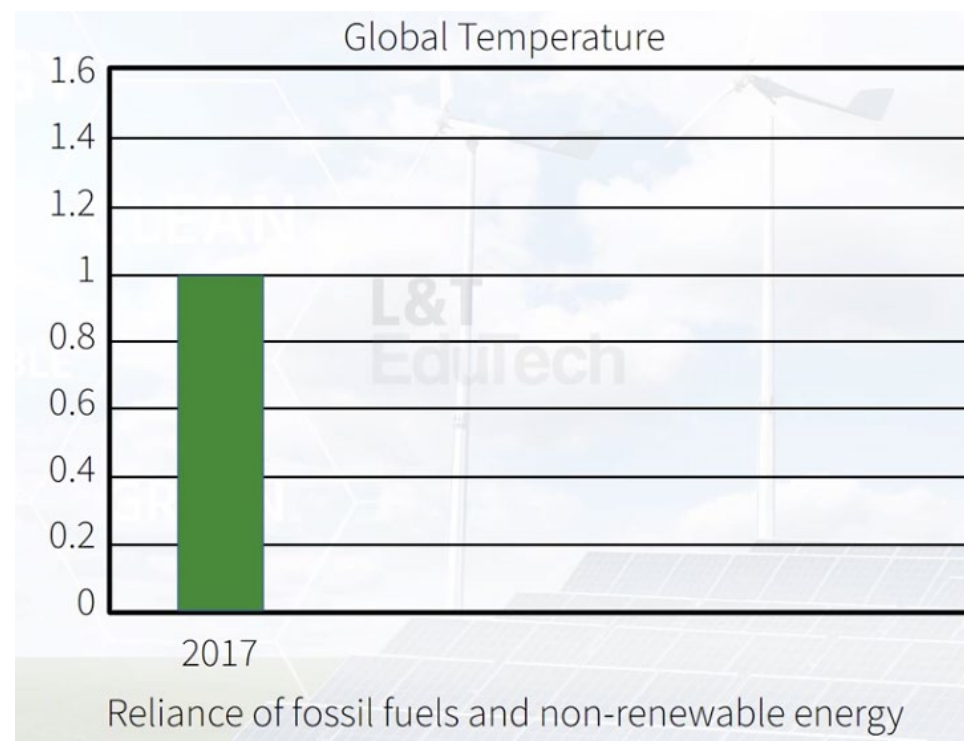
https://climate.ec.europa.eu/climate-change-explained/consequences-climate-change_en

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1.1 Introduction

Global Warming

As per scientists, the present reliance on fossil fuels and non-renewable energy has resulted in global temperatures rising one degree centigrade above the pre-industrial levels in 2017.

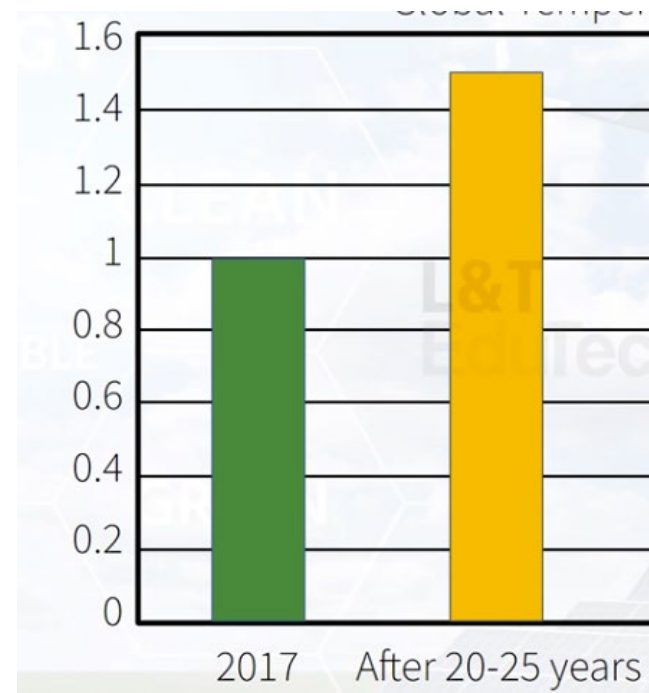


1. Photovoltaic Systems

1.1 Introduction

Global Warming

If we continue our reliance on fossil-based fuels, scientists believe the global temperatures would increase by 0.2 degrees centigrade every decade. The implications are that in the next 20-25 years, the global weather would be warmer by another 1.5 degree centigrade.



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1.1 Introduction

Global Warming

The idea is to keep global warming at the same levels and ensure there is no increase beyond the 1.5 degree centigrade as per the Paris Agreement and greenhouse gas emissions to be reduced by 45% by 2030 and reach net zero by 2050.



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1.1 Introduction

What are the impacts of the global warming on various climates?

Global warming has had an impact on melting of ice in the Arctic and Antarctic regions. Traditionally, these regions have cycles of summer and winter seasons wherein ice melts during summer while new snow is formed during winter.



1. Photovoltaic Systems

1.1 Introduction



What are the impacts of the global warming on various climates?

However, the greenhouse effect and global warming have skewed the weather pattern. Rising global temperatures have resulted in excessive melting of ice and reduced the formation of new snow. This excessive melting of ice has resulted in increased sea levels across the globe.



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What are the impacts of the global warming on various climates?

The rising sea levels now pose a threat to human habitat, especially those near and around coastal cities.



Rising sea levels will submerge large land mass, especially those in coastal areas, displacing hundreds of families and resulting in loss of infrastructure and livelihoods

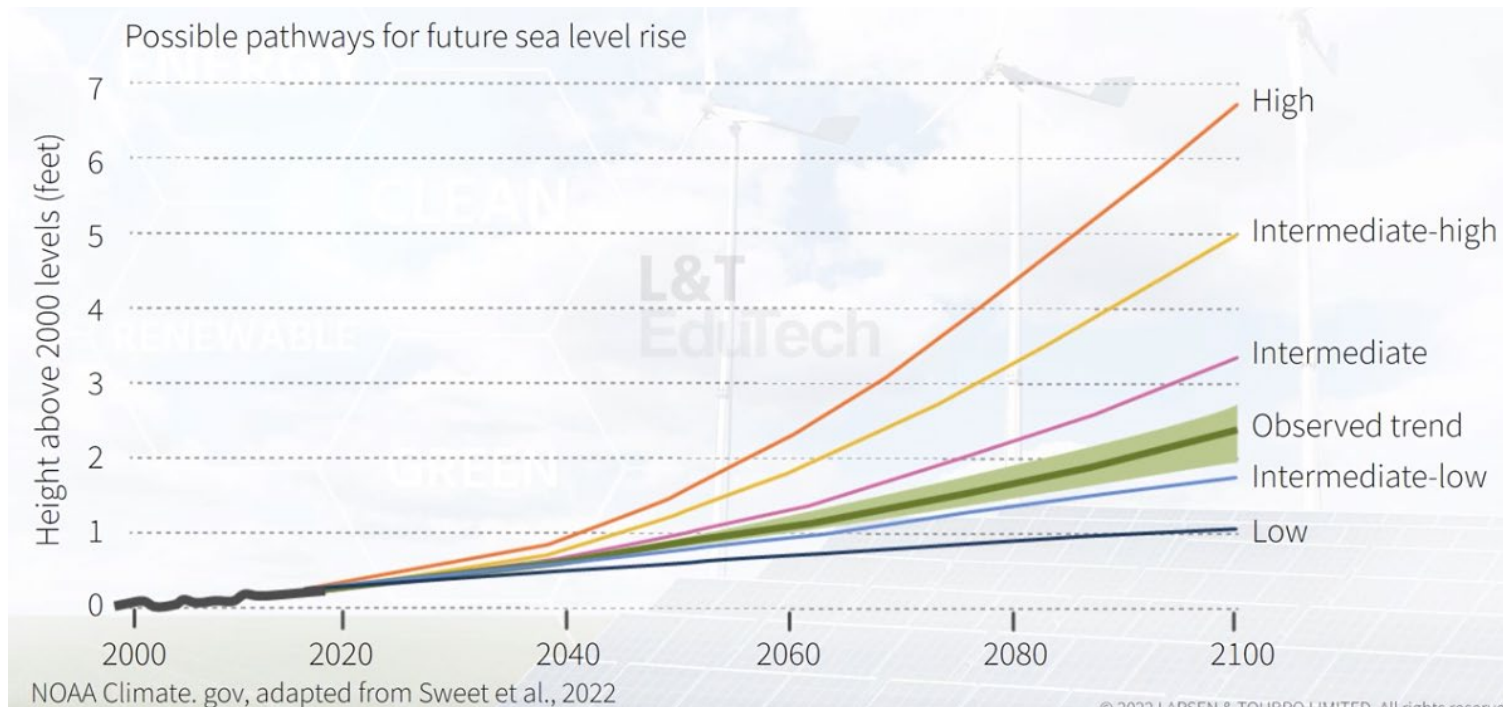
1. Photovoltaic Systems

1.1 Introduction



What are the impacts of the global warming on various climates?

An increase in sea levels can also result in the extinction of certain marine creatures, which, in turn, can impact the livelihood of those depending on the sea for subsistence. However, the impact of rising sea levels is not the same across the globe due to different coastlines.



<https://www.climate.gov/news-features/understanding-climate/climate-change-global-sea-level>

1. Photovoltaic Systems

1.1 Introduction

What are the impacts of the global warming on various climates?

Look at the impact of rising sea levels on major cities across the globe



1. Photovoltaic Systems

1.1 Introduction

What are the impacts of the global warming on various climates?
Mumbai Flood 2022

<https://indianexpress.com/article/cities/mumbai/mumbai-sea-bmc-risk-check-10866322/>

<https://www.bbc.com/news/world-asia-india-62056709>



Impact of Global Warming on Climate





Mumbai - 2022

Heavy rain lashes Mumbai

Flood in low-lying areas

1. Photovoltaic Systems

1.1 Introduction

What are the impacts of the global warming on various climates?

New York Flood 2021

<https://indianexpress.com/article/cities/mumbai/mumbai-sea-bmc-risk-check-10866322/>

<https://www.bbc.com/news/world-asia-india-62056709>



Impact of Global Warming on Climate



New York - 2021

New York flood killed 43 people

1. Photovoltaic Systems

1.1 Introduction



What are the impacts of the global warming on various climates?
Petropolis, Brazil 2022



Impact of Global Warming on Climate



Petropolis - 2022

Mudslides resulted in extensive
damage and loss of life

<https://floodlist.com/america/petropolis-floods-landslides-february-2022-update>

1. Photovoltaic Systems

1.1 Introduction



What are the impacts of the global warming on various climates?
Tohoku and Hokuriku, Japan 2022



Impact of Global Warming on Climate



Tohoku and Hokuriku regions - 2022

Around 5,40,000 people were told to evacuate after a heavy flood in Japan

https://pbv.or.jp/en/disaster_relief/domestic/



What are the impacts of the global warming on various climates?

IPCC

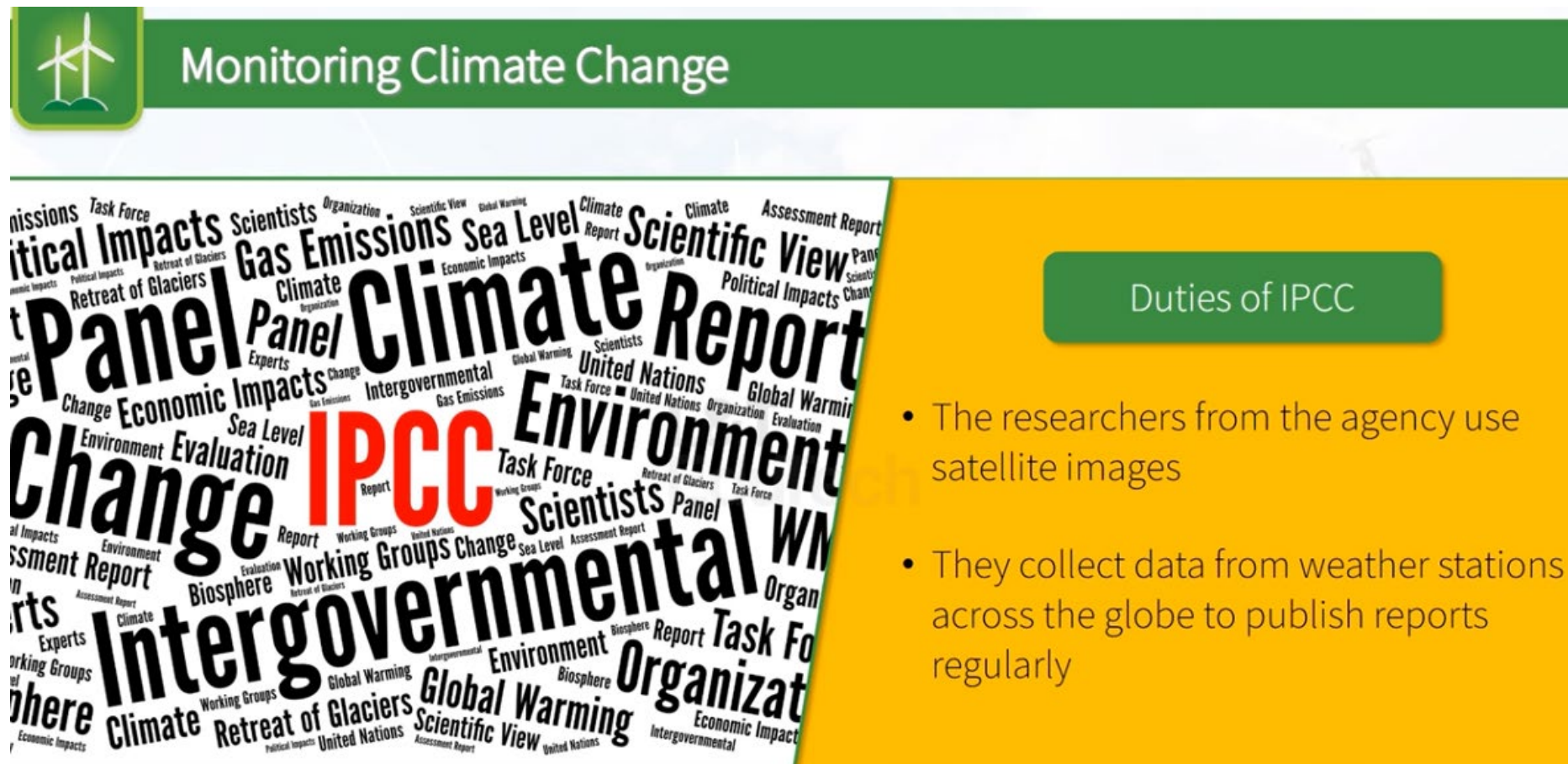


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What are the impacts of the global warming on various climates?

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What are the impacts of the global warming on various climates?

IPCC



Report on Global Warming



IPCC REPORT

40 – 45 %

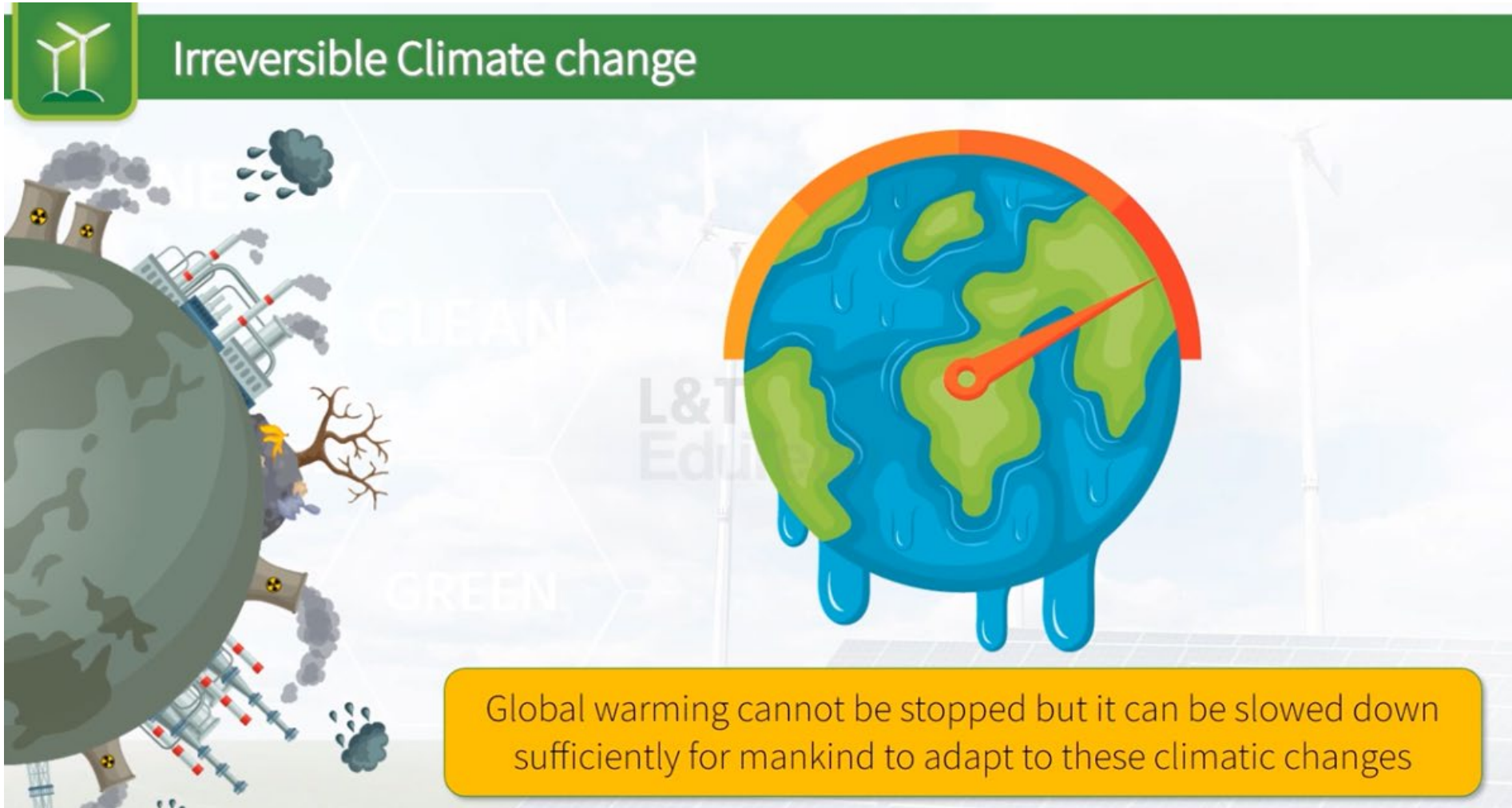
- Almost 3.3 billion people, or 40 to 45% of the global population, is set to face severe health challenges in the next couple of decades that is expected to increase deaths across vulnerable communities



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What are the impacts of the global warming on various climates?

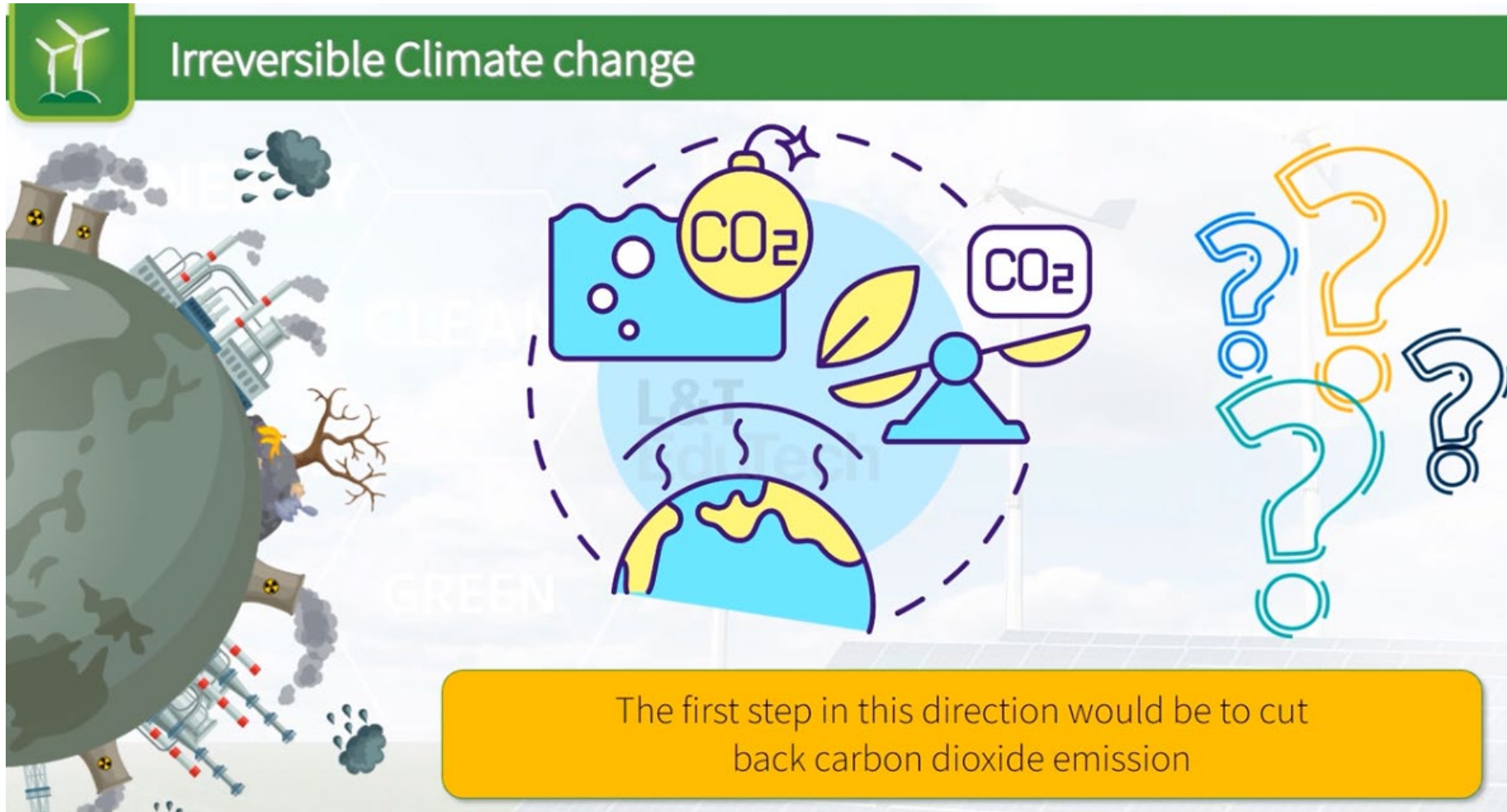




1. Photovoltaic Systems

1.1 Introduction

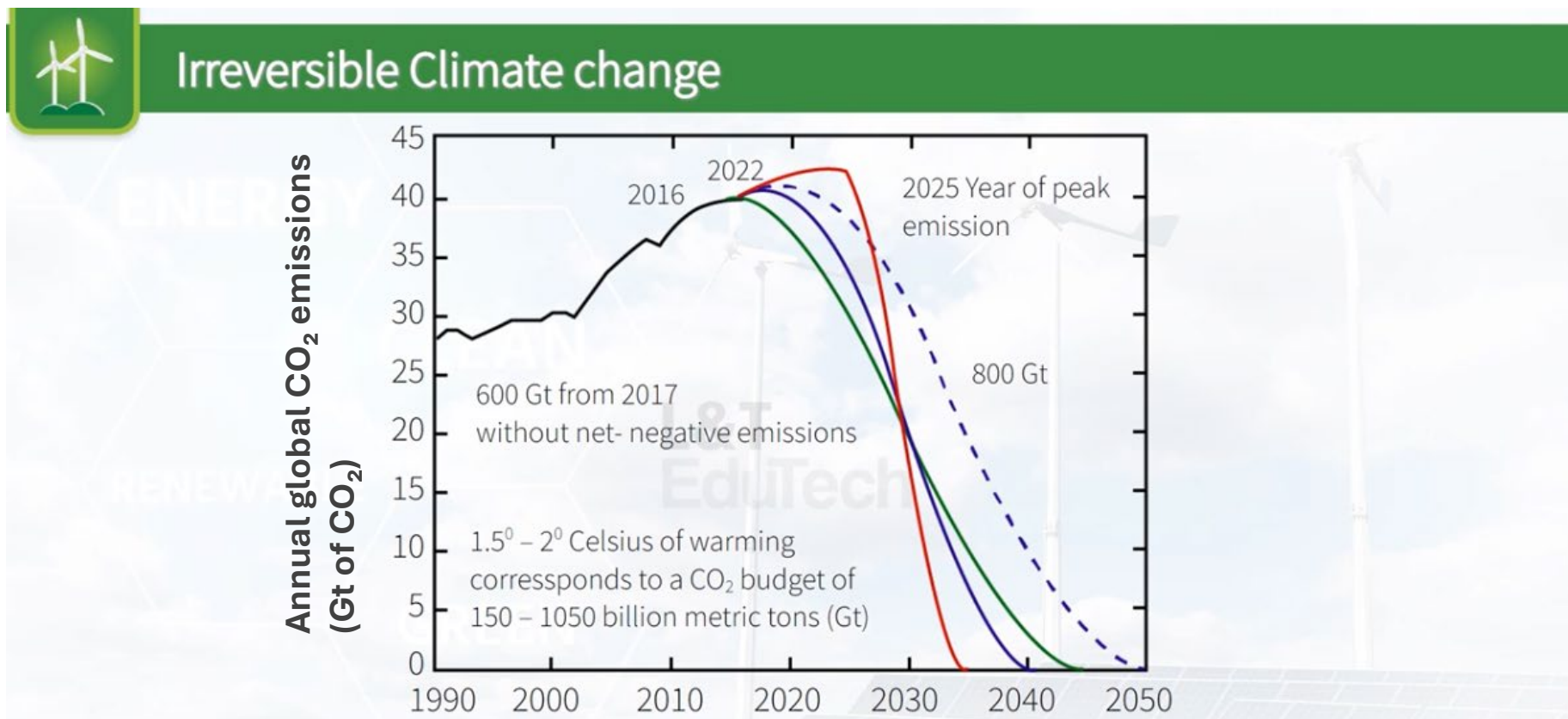
What are the impacts of the global warming on various climates?



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1.1 Introduction

What are the impacts of the global warming on various climates?



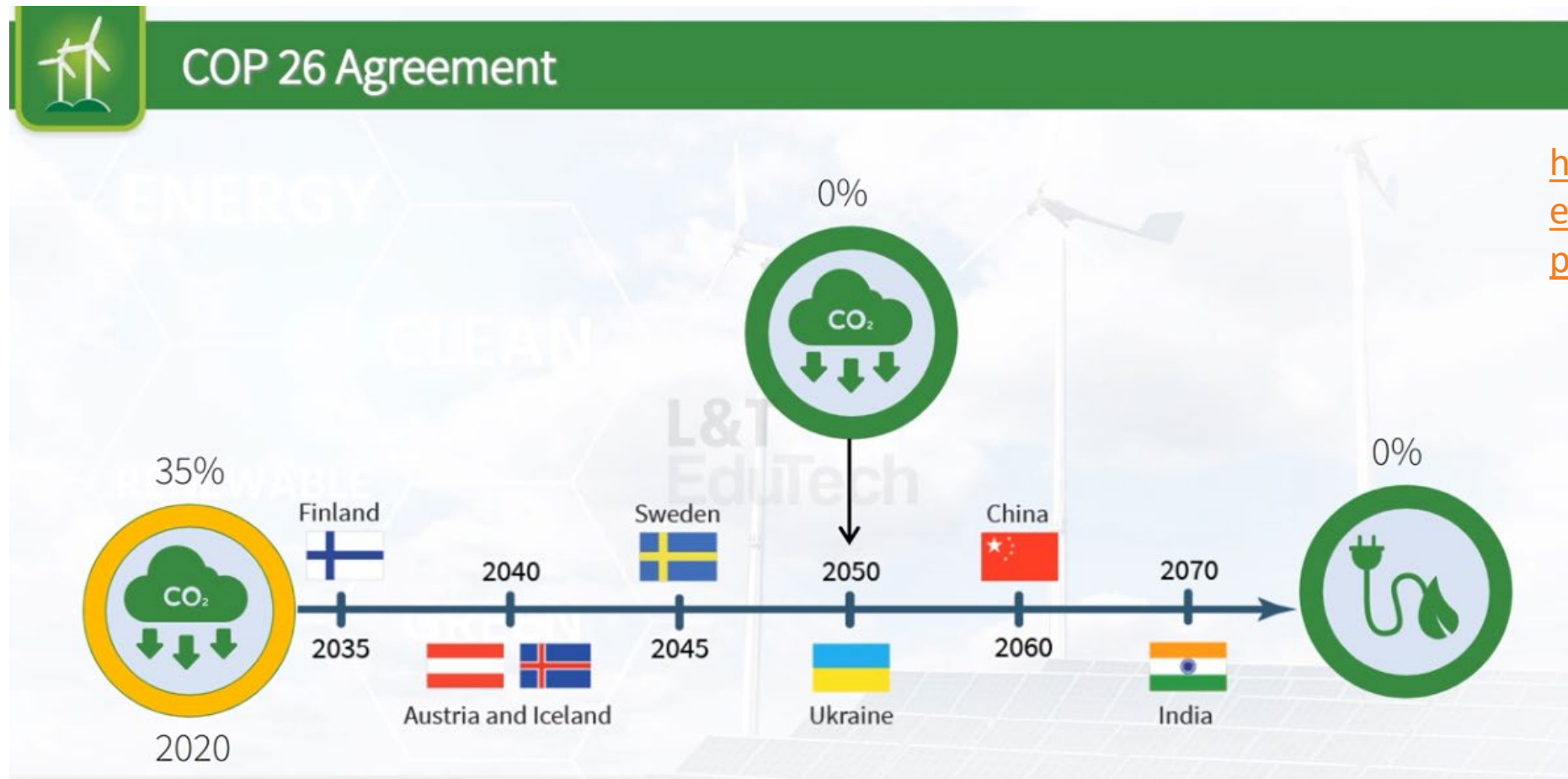
The carbon budget is the quantity of carbon dioxide which humanity can emit while still containing the global warming within 1.5 degree centigrade compared with pre-industrial levels, as per the Paris Agreement

1. Photovoltaic Systems

1.1 Introduction



What are the impacts of the global warming on various climates?



<https://www.un.org/en/climatechange/cop26>



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What are the impacts of the global warming on various climates?



Proactive Measures



Trucks



Shipping



Aviation

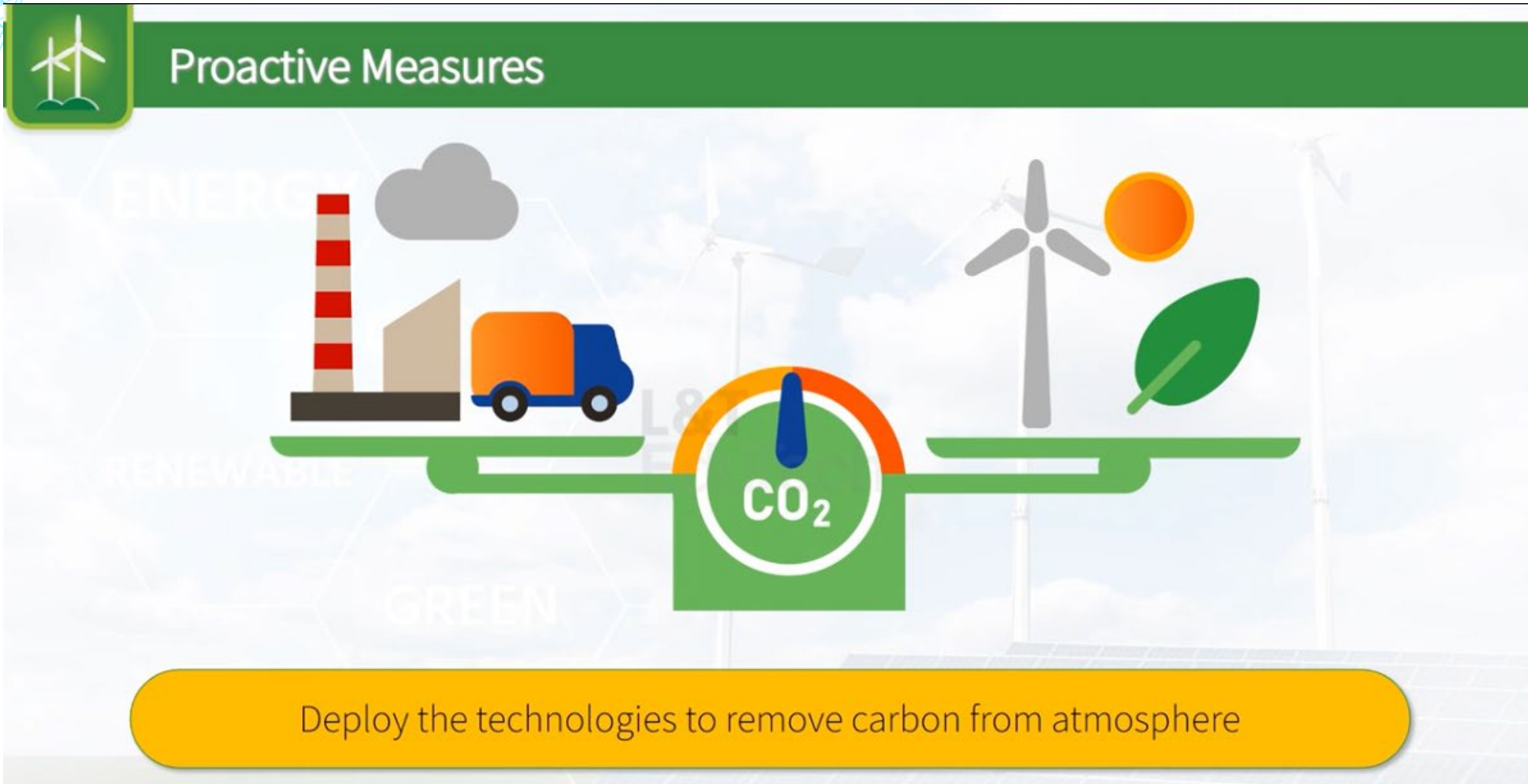
There are no holistic technological solutions that ensure zero emission at the source

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What are the impacts of the global warming on various climates?



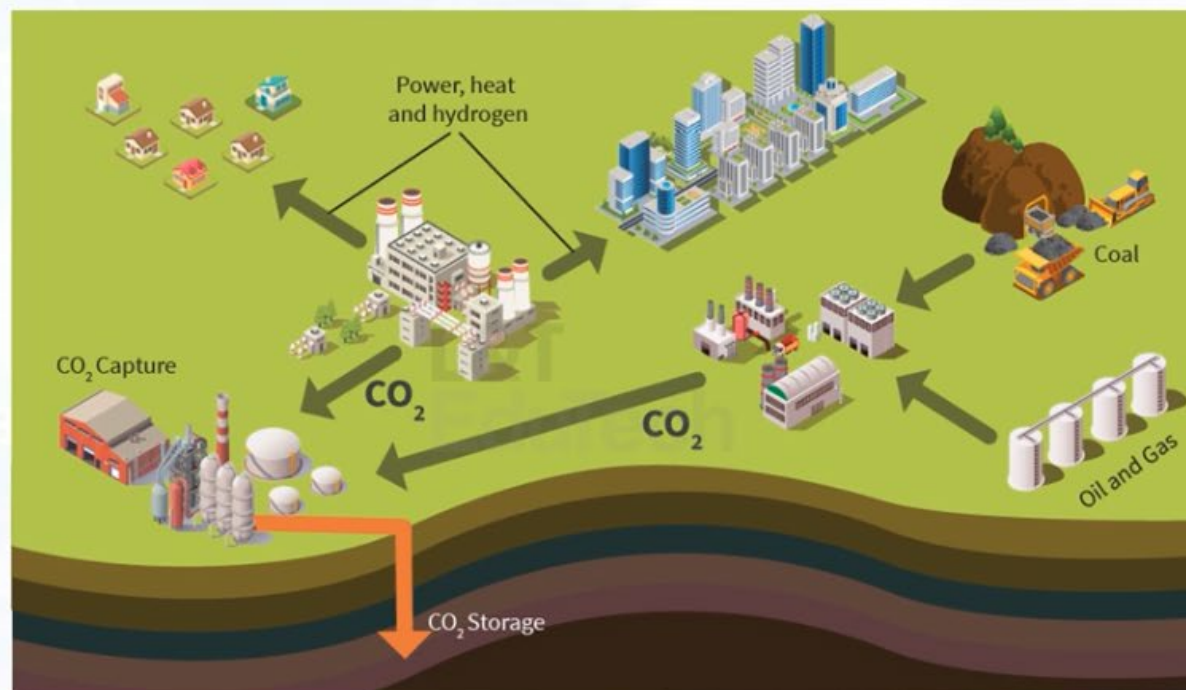
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What are the impacts of the global warming on various climates?



Proactive Measures



CCS is a technology that requires capturing the CO₂ produced by industrial activity and then transporting and storing it deep underground

1. Photovoltaic Systems

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What are the impacts of the global warming on various climates?



The illustration depicts a green banner at the top with a wind turbine icon and the text "Measures to Restore System". Below the banner, a yellow electric car is shown charging at a red station with a lightning bolt symbol, set against a background of green trees and a blue city skyline. To the right, a blue car is shown with four circular icons of people around it, representing a ride-sharing service. A red location pin icon is also present. The word "ENERGY" is faintly visible in the background. At the bottom, a yellow rounded rectangle contains the text: "Large-scale use of electric vehicles and ride-sharing services cuts down CO₂ emission and reduce the need for use of personal vehicles".

Measures to Restore System

ENERGY

Large-scale use of electric vehicles and ride-sharing services cuts down CO₂ emission and reduce the need for use of personal vehicles